

Calculations				
	Sheep	Pig	Cow	Goat
Meat Diet Portion	2.23%	7.21%	6.87%	1.89%
Meat MCal Target	61,082.39	197,396.75	188,030.12	51,715.76
Potential Dairy Value attached to each juvenile	132.54	0.00	1015.20	126.90
Value per juvenile	48.00	160.00	640.00	48.00
Adjusted Dairy Value attached to juvenile based on Dairy Utilization	66.27	0.00	507.60	63.45
Culled Mothers Value	14.77	4.41	105.26	14.77
Culled Adult Males	0.74	0.22	5.26	0.74
Final Value per juvenile	63.51	164.63	750.53	63.51
Value MCals	63.51	164.63	750.53	63.51
Offspring required Year ⁻¹	961.81	1,199.06	250.53	814.32
MCals from meat	61082.39	197396.75	188030.12	51715.76
Max Utilized Mcals allowed from Dairy	63739.20	0.00	127169.54	51668.77
MCals actually Utilized from Dairy	63739.20	0.00	127169.54	51668.77
Percent of diet from their meat	2.23%	7.21%	6.87%	1.89%
Percent of Diet from their Dairy	2.33%	0.00%	4.65%	1.89%
Kg Fleece	7,268.92	0.00	0.00	0.00
1 Year+ (age) skins	357.68	45.87	48.28	302.87
Yearly Offspring skins	1,750.41	1,413.16	369.32	1,482.09
Birthing Events per year	1.00	1.86	1.00	1.00
Offspring Mother ⁻¹ Year ⁻¹	0.56	5.48	0.68	0.56
Mothers Step 1	1,717.52	219.01	368.43	1,454.15
Males Step 1	480.91	599.53	125.27	407.16
Females Step 1	480.91	599.53	125.27	407.16
Replacement Mothers per year	343.80	44.10	46.24	291.13
Breeding Males Required	90.17	11.50	19.34	76.34
Additional Mothers	1,408.21	39.11	174.69	1,192.43
Males Step 2	394.30	107.05	59.39	333.88
Females 2	394.30	107.05	59.39	333.88
Total Mothers	2,061.32	263.11	414.67	1,745.28
Total Males	875.20	706.58	184.66	741.04
Total Females	86.61	492.48	65.87	73.28
Total Offspring	1,750.41	1,413.16	369.32	1,482.09
Total Animals	3,113.30	1,473.66	684.54	2,635.94
Units per Hectare	1.00			
Total Adults per Hectare	1.84	0.28	0.45	1.84
Total Offspring per Hectare	0.90	1.24	0.27	0.90
	1.53%		4.32%	1.30%
Total Adults	3,560	314	609	3,014
Total Offspring	1,750	1,413	369	1,482
Days for Maintenance	5.00	220.00	0.00	5.00
Dry Mothers	310.91	0.00	45.35	263.19
Other	28,006.54	0.00	0.00	23,713.37
MCals Dry Mothers	363,148.13	0.00	96,006.36	307,407.74
total Maintenance MCals	391,154.67	0.00	96,006.36	331,121.11

MCals Required				
	Sheep	Pig	Cow	Goat
Gestating	1,042,368	645,180	618,240	882,582
Lactating	1,988,639	229,356	664,774	1,683,797
Maintenance	391,155	0	96,006	331,121
Adult Males	105,318	0	40,948	89,168
Offspring	480,487	1,420,229	1,351,707	406,833
TOTAL	4,007,967.8	2,294,764.8	2,771,674.4	3,393,501.4

Animal Diet Breakdown				
	Sheep	Pig	Cow	Goat
Crop	Barley	Barley	Barley	Barley
Grams ⁻¹ Straw/Chaff	1000.0	0.0	4956.0	1000.0
MCal kg ⁻¹				
Grams ⁻¹ Grain	750.0	750.0	1500.0	750.0
MCal kg ⁻¹				
Days	90.0	90.0	90.0	90.0
Number of Animals	1,750.4	1,413.2	369.3	1,482.1
MCals from Straw/Chaff	223702.2	0.0	233917.9	189410.5
MCals from Grain	354457.8	286165.5	149574.1	300122.3
MCals from other feed	578160.0	286165.5	383492.0	489532.9

Diet Calculations		
Plants Mcal	1,947,422.483	
Meat Mcal	547,500	
Dairy	242,577.51729406	
	Perfect	
Plants %	71.14%	
Meat %	20.00%	
Dairy %	8.86%	
check	Perfect	
Percent		
Domestic Meat	18.20%	
Dairy	8.86%	
Wild Meat	1.80%	
Crops	64.02%	
Wild Plants	7.11%	
check	Perfect	
new dairy	242577.52	
potential dairy < limit	TRUE	
new limit	80.0%	
the limit greater than that left to satisfy	TRUE	
limit	1	
limit dairy checked	FALSE	
Mcalories		Surplus
Domestic Meat	498,225.000	
Dairy	242,577.517	
Wild Meat	49275.000	
Crops	1,752,680.234	
Wild Plants	194,742.248	
check	Perfect	
SURPLUS DAIRY	0.000	
This could account for additional % of diet	0.00%	

Herd Modelling

	Sheep	Pig	Cow	Goat				
Mothers Req'd	2061	263	415	1745				
Males for Slaughter	875	707	185	741				
Females for Slaughter	87	492	66	73				
Offspring available for Consumption	1750	1413	369	1482				
Gestation Req'd	572758	547431	419389	484929				
Lactation Req'd	1092713	194607	450956	925152				
Maintenance Req'd	1299619	0	123706	1100384				
Adult Males	105318	0	40948	89168				
Offspring Required	575009	1547413	1617616	486865				
Energy Required	3645419	2289451	2652615	3086499				
Ewe Fleece Kg Yr-1	4638							
Ram Fleece Kg year-1	180							
Juvenile Fleece Kg Yr-1	2451							
Kg Fleece Year-1	7269	0	0	0				